

TOPIC 5: BASIC COMPUTER HARDWARE COMPONENTS II – STORAGE UNITS

1. Meaning of Storage

Storage is the part of the computer that holds data and programs permanently or temporarily so they can be retrieved when needed.

2. Primary Storage

Also called main memory. It is fast but usually temporary (volatile).

- **RAM (Random Access Memory)** – temporary working memory; data is lost when power is off.
- **ROM (Read Only Memory)** – permanent memory that stores start-up instructions; cannot be easily changed.

3. Secondary Storage

Long-term storage that retains data even when the computer is off.

- Hard Disk Drive (HDD) – magnetic storage inside the computer
- Solid State Drive (SSD) – faster, no moving parts
- Flash drive / USB stick – portable
- Memory card (SD card)
- Optical discs (CD, DVD, Blu-ray)
- Cloud storage (online)

4. Storage Measurements

- **Bit** – smallest unit (0 or 1)
- **Byte** – 8 bits (enough for one character)
- **Kilobyte (KB)** $\approx 1,024$ bytes
- **Megabyte (MB)** $\approx 1,024$ KB
- **Gigabyte (GB)** $\approx 1,024$ MB
- **Terabyte (TB)** $\approx 1,024$ GB

5. Importance of Data Storage

- Keeps programs and files for future use
- Allows backup of important information
- Enables sharing and transferring of data
- Supports the running of software and the operating system

6. Summary

Primary storage (RAM, ROM) is fast and mostly temporary; secondary storage (hard disk, flash drive, etc.) is permanent. Storage capacity is measured in bits, bytes, KB, MB, GB and TB.

End of Topic 5